

Software Engineering (ENCT 352): Chapter-wise Past Questions

Source: TU IOE BCT III/I Software Engineering (CT 601) papers, **2068 Chaitra to 2083 Baishakh** (27 papers, regular and back), sorted into the chapters of the new **ENCT 352** syllabus.

How to read this

- Similar questions from different years are combined into one entry. The years it was asked are in square brackets at the end.
- ★ = asked 5 or more times. Study these first.
- Case-study DFD, use-case and FR/NFR questions are listed with their scenario so you can practise them.
- Old CT 601 topics that are **not** in ENCT 352 (real-time systems, software reuse/CBSE, COCOMO/FP estimation) are in the **Appendix** at the end. Check with your teacher whether they can still come.

Chapter 1: Introduction

1. ★ What is software? What are the typical characteristics / qualities of good software? **[2081 Bhadra, 2078 Bhadra, 2078 Kartik, 2076 Chaitra, 2073 Chaitra, 2072 Kartik]**
2. ★ Justify the statement "Software doesn't wear out (like hardware components)." **[2083 Baishakh, 2082 Bhadra, 2082 Baishakh, 2081 Bhadra, 2078 Bhadra]**
3. ★ What is software crisis? Explain with an example / with its history. **[2083 Baishakh, 2079 Bhadra, 2078 Kartik, 2076 Chaitra, 2074 Ashwin, 2072 Kartik, 2068 Chaitra]**
4. How have modern software engineering practices helped overcome the software crisis? **[2082 Bhadra]**
5. Define software crisis. How can you say that there was a software crisis in the late 60s? **[2075 Chaitra]**
6. What factors have contributed to the present software crisis? Suggest possible solutions. **[2075 Ashwin]**
7. What is software crisis and what are its reasons? Describe the evolutionary model and how it reduces the crisis problem. **[2073 Shrawan]**
8. What are the reasons for software runaways? **[2072 Kartik]**
9. What makes the software development process complex? Why do simple man-month measurement and adding more workers to a delayed project not work? **[2070 Ashad]**

No past questions yet on: application domains, legacy software, software myths, essence of practice / general principles. These are new in ENCT 352, so prepare short notes on them.

Chapter 2: The Software Process

Process models in general

1. What is a software process model? Why are different process models used in software development? **[2080 Bhadra, 2080 Baishakh, 2078 Kartik, 2069 Ashad]**
2. Explain the fundamental activities of the software process. **[2082 Bhadra]**
3. Describe the main activities in the software design process and the output of these activities. **[2069 Ashad]**
4. Explain the current model of the software process. **[2071 Chaitra]**

Waterfall

5. What is SDLC? Discuss the stages of the waterfall model. What are its demerits / advantages and disadvantages? **[2081 Baishakh, 2079 Bhadra]**
6. Explain the waterfall model. Why is it not suitable when important functionality must be delivered in a short time? **[2073 Chaitra]**
7. Why is the waterfall model not an accurate reflection of software development activities? Explain a better alternative model. **[2071 Shrawan, 2071 Chaitra]**

Incremental

8. ★ Explain the incremental model with its characteristics, advantages and disadvantages. **[2081 Bhadra, 2076 Ashwin, 2075 Chaitra, 2080 Baishakh, 2082 Bhadra]**
9. You are leading a team building an e-commerce site using the incremental model. How do you implement it? **[2082 Bhadra]**
10. Differentiate the spiral model and the incremental model, with advantages and disadvantages. **[2080 Baishakh]**

Spiral / prototyping / evolutionary

11. ★ Explain the spiral model with advantages and disadvantages. Why is it preferred in high-risk projects? **[2083 Baishakh, 2080 Bhadra, 2068 Chaitra, 2080 Baishakh, 2074 Chaitra]**
12. How can both the waterfall model and the prototyping model be accommodated in the spiral model? **[2078 Kartik, 2072 Kartik]**
13. A travel agency wants to automate simple bookkeeping and is unsure about the user interface. Would the spiral model be proper? Justify. **[2075 Ashwin]**
14. A client's problem has uncertainties that lead to loss if not planned. Which model do you suggest? Justify, and explain its pros and cons. **[2074 Chaitra]**
15. Explain the prototyping model with its advantages and disadvantages. In what types of project can it be used? **[2082 Baishakh, 2076 Chaitra]**
16. What is a prototype? What are the risks if prototyping becomes uncontrolled? Explain RAD in brief. **[2072 Chaitra]**
17. Describe the evolutionary model and how it reduces the crisis problem. **[2073 Shrawan]**
18. Short note: Risk analysis. **[2070 Chaitra]**

No past questions yet on Agile (manifesto, 12 principles, Scrum, XP, Lean, agile vs plan-driven, model selection). All of it is new in ENCT 352, so expect questions here.

Chapter 3: Software Requirements Engineering

Feasibility

1. ★ Why do we need to do a feasibility study before accepting a project? What are its types? Explain any two. **[2083 Baishakh, 2082 Bhadra, 2080 Bhadra, 2069 Chaitra, 2068 Chaitra (short note)]**
2. Discuss all the activities in problem definition and feasibility analysis. **[2072 Chaitra]**
3. What are the major components of a feasibility study report? Justify why the candidate matrix with recommendation is considered standard. **[2070 Chaitra]**
4. Prepare a project feasibility report (telemedicine project / GPS public transport system). **[2074 Chaitra, 2073 Chaitra]**

RE process, elicitation, analysis

5. Why do we need requirement analysis and refinement during requirement engineering? **[2083 Baishakh]**
6. What is requirement engineering? Explain the RE process / its steps in detail. **[2078 Bhadra, 2074 Chaitra]**
7. "Requirements engineering is the process of identifying, eliciting, analysing, specifying, validating and managing stakeholder needs." Justify with examples. **[2080 Bhadra]**
8. Why is it so difficult to gain a clear understanding of what the customer wants? Describe guidelines for the requirement elicitation process. **[2081 Bhadra, 2075 Ashwin, 2071 Chaitra]**
9. What do you mean by requirement elicitation and analysis? **[2076 Ashwin, 2069 Ashad, 2074 Ashwin (short note)]**
10. What are the techniques for requirement gathering and analysis? Explain at least three methods, with a comparison chart of pros and cons. **[2070 Ashad, 2070 Chaitra]**
11. Explain the requirement management process with an illustration. **[2068 Chaitra]**
12. Short note: Requirements discovery through prototyping. **[2073 Chaitra]**
13. Compare consistency and completeness in requirements engineering. **[2072 Chaitra]**
14. "Walking on water and developing software from a specification are easy if both are frozen." Justify. **[2074 Chaitra]**

SRS and requirement types

15. ★ What is the importance and use of the SRS / software requirements document? **[2082 Baishakh, 2080 Bhadra, 2079 Bhadra, 2078 Bhadra, 2071 Chaitra (short note)]**
16. Why is the SRS required during development? Justify the extra effort spent on detailed specification early. **[2069 Chaitra]**
17. Give your view on requirement engineering and requirement specification. **[2071 Shrawan]**
18. ★ Differentiate functional and non-functional requirements with examples. **[2083 Baishakh, 2076 Ashwin, 2069 Chaitra, 2069 Ashad, 2073 Shrawan (short note on NFR)]**
19. Distinguish user requirements and system requirements. Why must an engineer ensure both FR and NFR are in the specification? **[2079 Bhadra, 2074 Ashwin]**

Case study: list the FR / NFR for

20. Movie theatre online ticket booking (FR only). **[2082 Bhadra]**
21. Airline reservation system. **[2082 Baishakh]**
22. Online food ordering system (order food and make payment). **[2081 Bhadra]**
23. Automated bus-ticket issuing machine (NFR, with use case). **[2081 Baishakh]**
24. Online examination web app. **[2080 Bhadra]**

25. Telemedicine project of Ministry of Health. **[2080 Baishakh, 2074 Chaitra]**
26. Issuing a book from a library. **[2079 Bhadra]**
27. Restaurant information system (orders, kitchen, inventory, reports). **[2078 Kartik, 2068 Chaitra]**
28. Doctor who sees patients and prescribes medicine (with use case). **[2078 Bhadra]**
29. Real-time ride-sharing system. **[2076 Chaitra]**
30. Time Management System (TMS) for executives. **[2075 Ashwin]**
31. Mahanpur Nagarpalika tourist system (quality attributes incl. FR and NFR). **[2074 Ashwin]**

No past questions yet on the agile RE topics (user stories, acceptance criteria, product backlog, story mapping, backlog grooming).

Chapter 4: Architectural Design

1. ★ What is architectural design / software architecture? Why is architecture important to drive software development? **[2083 Baishakh, 2082 Baishakh, 2081 Bhadra, 2081 Baishakh, 2080 Bhadra, 2080 Baishakh, 2078 Bhadra, 2076 Chaitra, 2076 Ashwin, 2075 Chaitra, 2074 Ashwin, 2069 Ashad, 2068 Chaitra]**
2. Why is it necessary to design the system architecture before the specification is written? **[2075 Ashwin, 2073 Chaitra, 2071 Chaitra]**
3. What is the role of reference architectures in system design? Justify with an example. **[2069 Chaitra]**
4. Explain 2-tier, 3-tier and multi-tier architecture with examples. **[2083 Baishakh, 2078 Bhadra, 2074 Ashwin]**
5. ★ Explain client-server architecture and its importance, with an example. **[2082 Baishakh, 2076 Chaitra, 2071 Chaitra, 2081 Baishakh, 2074 Ashwin (compare)]**
6. Differentiate thin client and thick (fat) client models. **[2082 Bhadra, 2081 Baishakh, 2073 Shrawan, 2073 Chaitra]**
7. Describe the layered architecture / layered model with an example. **[2082 Bhadra, 2081 Baishakh, 2080 Bhadra]**
8. What is the repository model? Differentiate it from the layered model with advantages and disadvantages. **[2080 Bhadra, 2079 Bhadra]**
9. Explain pipe-and-filter architecture and call-return architecture. **[2080 Baishakh, 2078 Kartik]**
10. What are control styles in architectural design? Explain the types with examples. **[2078 Kartik, 2076 Ashwin, 2070 Ashad, 2069 Ashad]**
11. ★ What are modular decomposition styles? How is modular decomposition practised in system design? **[2081 Bhadra, 2079 Bhadra, 2075 Chaitra, 2075 Ashwin, 2071 Shrawan, 2076 Chaitra (short note), 2074 Ashwin (short note), 2074 Chaitra (short note)]**
12. Describe multiprocessor architecture with an example. **[2073 Shrawan, 2068 Chaitra]**
13. Explain distributed object architecture with an example. **[2073 Chaitra, 2074 Chaitra (short note)]**
14. Compare:
 - Client-server vs distributed object architecture **[2074 Ashwin, 2072 Kartik]**
 - Multiprocessor vs client-server architecture **[2070 Ashad]**
 - Distributed object vs multiprocessor architecture **[2069 Chaitra]**

15. Why do we prefer a lower value of fan-out / fan-in for a function as a software metric? (coupling) **[2083 Baishakh, 2082 Bhadra]**

Cohesion and coupling are in the new syllabus (4.4). The only related past question is Q15, so prepare them separately.

Chapter 5: System Modeling

General

1. Why is system modeling important? What are the weaknesses of the structured analysis method? **[2068 Chaitra]**
2. What is a behavior model? How does it differ from a data model of the same system? **[2072 Kartik, 2074 Ashwin (process model vs data model)]**
3. What is behavior modeling in systems analysis? Illustrate with a model of a web-based transaction portal. **[2071 Shrawan]**
4. Explain the object aggregation model with an example. **[2069 Ashad]**
5. Short note: Object oriented analysis. **[2070 Chaitra]**
6. Compare black hole and miracle in DFD. **[2072 Chaitra]**
7. What is DFD balancing? **[2071 Chaitra]**

DFD case studies (level 0 / context, level 1, and sometimes level 2)

8. Bus garage repair automation system. **[2083 Baishakh]**
9. TU Examination Management System (EMS). **[2082 Bhadra]**
10. Student Attendance Management System. **[2082 Baishakh]**
11. Travel agency Airline Ticketing System. **[2081 Bhadra, 2075 Chaitra]**
12. Online food ordering system (context and level 1). **[2081 Baishakh]**
13. Online examination web app. **[2080 Bhadra]**
14. Telemedicine project (level 1). **[2080 Baishakh]**
15. Restaurant order / inventory / vendor system (context DFD + use case). **[2080 Baishakh]**
16. Online shopping system. **[2079 Bhadra]**
17. Restaurant information system. **[2078 Kartik]**
18. Real-time ride-sharing system. **[2076 Chaitra]**
19. College sports week (list processes and agents, DFD up to level 1). **[2076 Ashwin]**
20. Time Management System (context diagram up to level 2). **[2075 Ashwin]**
21. Bank cheque cash-payment process (physical and logical DFD). **[2074 Chaitra]**
22. School departments and graduation (processes and agents, DFD, relationship between instructor, HOD and department). **[2073 Shrawan]**
23. GPS public transport with smart card payment (context + two level-2 DFDs). **[2073 Chaitra]**
24. E-commerce order processing (two DFDs). **[2072 Chaitra]**
25. Online DVD movie portal (level 1). **[2071 Chaitra]**
26. Any example of a second-level DFD showing modular decomposition. **[2071 Shrawan]**

Use case diagrams

27. Online food delivery with Fonepay payment. **[2083 Baishakh]**

28. Event management system (conferences, weddings, concerts). **[2082 Bhadra]**
29. Online food ordering app. **[2082 Baishakh]**
30. Online Jyotis (astrologer) appointment booking app. **[2081 Bhadra]**
31. Automated bus-ticket issuing machine. **[2081 Baishakh]**
32. Restaurant system. **[2080 Baishakh]**
33. Doctor who sees patients and prescribes medicine. **[2078 Bhadra]**

Data / behavior models for a scenario

34. Mahanpur Nagarpalika tourist system: complete data models. **[2074 Ashwin]**
35. Mero College class scheduling: system-level data model. **[2070 Ashad]**
36. Bhrantipur Book Store: system-level process (behavior) model. **[2069 Chaitra]**

No past questions yet on activity diagrams, class diagrams or use-case descriptions as separate topics. They are in the new syllabus (5.3, 5.4).

Chapter 6: Coding and Testing

Verification and validation

1. ★ Differentiate verification and validation, with examples. **[2083 Baishakh, 2082 Baishakh, 2081 Baishakh, 2080 Bhadra, 2080 Baishakh, 2079 Bhadra, 2078 Kartik, 2078 Bhadra, 2076 Chaitra, 2076 Ashwin, 2075 Chaitra, 2074 Chaitra, 2072 Kartik, 2070 Ashad, 2070 Chaitra, 2069 Ashad, 2075 Ashwin (short note)]**
2. What are the goals of V&V? Why is V&V / verification planning necessary, and what are its steps? **[2075 Chaitra, 2071 Shrawan, 2069 Ashad]**
3. Why is software verification essential before launching a system? What is its role in ensuring correct implementation? **[2074 Ashwin, 2070 Ashad]**
4. "Validation examines the dynamic behavior of a software system." Explain with an example. **[2073 Shrawan]**
5. What are critical systems? How does partitioning help in the V&V process? **[2072 Kartik]**
6. What type of systems are validated using formal methods, and why? What are formal methods of verification? **[2083 Baishakh, 2082 Bhadra, 2069 Ashad]**
7. Formal verification could be automated if the specification were complete and precise. Why is this hard in practice? **[2072 Chaitra]**
8. Explain the V-model and how it emphasises V&V. Discuss the hierarchical levels of testing. **[2078 Bhadra, 2076 Ashwin]**

Code review / inspection

9. ★ Explain the software inspection process. What faults can be uncovered by software inspection? **[2082 Bhadra, 2082 Baishakh, 2081 Bhadra, 2081 Baishakh, 2080 Bhadra, 2080 Baishakh, 2079 Bhadra, 2078 Kartik, 2076 Chaitra, 2076 Ashwin, 2069 Ashad]**
10. Why is program inspection an effective technique for finding errors? What errors are unlikely to be found by inspection? **[2068 Chaitra]**
11. Prepare a checklist for software code inspection. **[2073 Chaitra]**
12. Compare walkthrough and inspection. **[2072 Kartik]**

13. Short note: Cleanroom engineering. [2070 Chaitra]

Testing levels and types

14. ★ Differentiate alpha testing and beta testing, and their importance. [2082 Bhadra, 2081 Bhadra, 2076 Ashwin, 2074 Chaitra, 2073 Shrawan, 2073 Chaitra, 2072 Chaitra, 2070 Chaitra, 2069 Chaitra]
15. ★ Compare unit, integration and system testing. [2081 Baishakh, 2078 Kartik, 2074 Chaitra, 2073 Shrawan, 2073 Chaitra]
16. Compare integration testing and incremental integration testing. [2083 Baishakh]
17. Why is a unit test not enough for a complex system with many interacting units? What other tests are needed? [2070 Ashad]
18. What are the types of system testing? Explain the merits of object-oriented testing. [2069 Ashad]
19. Establish the chronology of component, release, unit and integration testing. [2071 Chaitra]
20. Write about stub and driver testing. [2075 Chaitra]
21. Define regression testing. [2078 Kartik]
22. What is the difference between testing and debugging? [2080 Bhadra]
23. What is exception and error testing in system implementation? [2071 Shrawan]
24. Develop a complete test strategy for the Time Management System and document it in a test specification. [2075 Ashwin]

Black-box and white-box

25. ★ Differentiate black-box and white-box testing. Are both necessary? [2082 Baishakh, 2080 Bhadra, 2079 Bhadra, 2076 Chaitra, 2075 Chaitra, 2074 Ashwin, 2073 Shrawan, 2069 Chaitra, 2072 Kartik (functional vs structural), 2070 Chaitra (short note)]
26. Why can't an end user perform white-box testing? [2083 Baishakh]
27. What is equivalence partitioning? How does it help in testing? Examples: campus roll number, telephone number. [2082 Bhadra, 2070 Ashad, 2069 Chaitra]
28. Design test cases for a program that finds the nature of the roots of a quadratic equation (a, b, c in [0, 100]). [2068 Chaitra]
29. ★ Explain cyclomatic complexity as a software metric, with an example. [2081 Baishakh, 2080 Baishakh, 2078 Kartik, 2075 Chaitra, 2072 Kartik]
30. Short note: Software testing metrics. [2071 Shrawan]

No past questions yet on coding standards, TDD, continuous testing, automated unit testing or refactoring. These are new in ENCT 352.

Chapter 7: Software Quality, Assurance, Maintenance

Quality concepts and attributes

1. Define software quality and explain the rationale for your definition. Explain quality attributes. [2073 Shrawan]
2. Distinguish external and internal software qualities. Differentiate correctness, reliability and robustness. [2081 Baishakh]

3. What are software quality measures? **[2074 Ashwin, 2074 Chaitra]**
4. Why are software quality standards needed? What is the importance of quality management? **[2075 Ashwin, 2072 Kartik]**
5. Explain Garvin's quality dimensions. **[2068 Chaitra]**
6. "Software quality, reliability and safety seem similar but are fundamentally different." Discuss. **[2069 Ashad]**
7. What is software reliability? How can we measure it? Why is it important? Give examples of disasters caused by software failure. **[2078 Kartik, 2076 Ashwin, 2075 Chaitra, 2070 Chaitra]**

SQA and reviews

8. ★ What is SQA? Why is it important, and how does it help enhance software quality? **[2083 Baishakh, 2080 Bhadra, 2079 Bhadra, 2076 Ashwin, 2075 Chaitra, 2074 Ashwin, 2070 Chaitra (short note), 2070 Ashad (short note), 2069 Chaitra (short note)]**
9. What is the significance of "the vital few" in statistical SQA? **[2083 Baishakh, 2082 Bhadra, 2081 Bhadra, 2071 Shrawan (short note)]**
10. Short note: SQA plan. **[2073 Chaitra]**
11. ★ What is FTR? How is a Formal Technical Review conducted? What are its objectives, and how does it help product quality? **[2082 Bhadra, 2081 Baishakh, 2080 Bhadra, 2080 Baishakh, 2078 Bhadra, 2076 Chaitra, 2076 Ashwin, 2075 Chaitra, 2068 Chaitra, 2075 Ashwin (short note), 2074 Chaitra (short note, FTR and inspection for QC)]**
12. Compare inspection and review in software quality management. **[2073 Chaitra]**
13. Compare traceability and adaptability in reviewing steps. **[2072 Chaitra]**

CMM / CMMI / ISO

14. ★ What is CMMI? Explain its levels / how it evaluates the maturity of software development. **[2082 Baishakh, 2081 Bhadra, 2080 Bhadra, 2079 Bhadra, 2078 Bhadra, 2076 Chaitra, 2074 Ashwin, 2070 Chaitra]**
15. Describe the staged CMMI model with levels and key process areas. **[2083 Baishakh, 2082 Bhadra, 2075 Ashwin, 2074 Chaitra]**
16. What are the characteristics of systems at different CMMI levels? Explain the two top levels. **[2069 Chaitra]**
17. ★ Explain CMM and its levels. Explain any three (or two) levels. What advantages does a company get by moving to the top level? **[2078 Kartik, 2073 Chaitra, 2072 Kartik, 2072 Chaitra, 2070 Ashad, 2076 Ashwin (short note: SEI CMM), 2071 Shrawan (short note: CMM level)]**
18. How does SEI CMM ensure quality? How does CMM encourage continuous process improvement? **[2076 Ashwin, 2071 Chaitra]**
19. What are the differences between CMM and ISO standards? **[2080 Baishakh, 2073 Chaitra, 2071 Chaitra, 2069 Ashad]**

Maintenance

20. Describe the different types of software maintenance. **[2070 Chaitra]**

Maintenance effort and lifecycle considerations are in the new syllabus (7.6) but have not been asked.

Chapter 8: Software Configuration Management

1. ★ Define version, variant and release in configuration management. Differentiate version and variant. **[2083 Baishakh, 2082 Bhadra, 2082 Baishakh, 2081 Bhadra, 2081 Baishakh, 2080 Baishakh, 2079 Bhadra, 2078 Bhadra, 2076 Ashwin (short note), 2073 Chaitra (short note)]**
2. What is the difference between version and release? Why do we need SCM? **[2073 Shrawan, 2076 Chaitra (short note: need of SCM)]**
3. Define configuration management. Explain SCM activities. **[2080 Bhadra]**
4. ★ Describe the change management process. **[2082 Baishakh, 2081 Bhadra, 2078 Bhadra, 2071 Chaitra (short note), 2068 Chaitra (short note)]**
5. Compare change management and version management. **[2074 Ashwin]**
6. Explain version naming approaches / the versioning process with its components. **[2080 Baishakh, 2071 Shrawan]**
7. What is the release problem? **[2082 Bhadra]**
8. Short note: Version and release management / version control. **[2078 Kartik, 2069 Chaitra, 2068 Chaitra]**
9. What are the main objectives of configuration management and version control? What are codeline and baseline? Compare them. **[2073 Chaitra, 2072 Chaitra]**
10. How is configuration management planning done? **[2075 Chaitra, 2069 Ashad]**
11. "Survival of the fittest is valid in today's software industry." Explain in the context of issues modern SCM must address. **[2072 Kartik]**

No past questions yet on continuous integration or branching strategies (Git-flow). These are new in ENCT 352.

Chapter 9: Recent Trends

No questions in any past paper. The whole chapter is new in ENCT 352: DevOps, CI/CD pipelines, cloud-native and microservices, infrastructure as code, AI-assisted development, low-code/no-code, green software engineering. Prepare short notes on each.

Appendix: Old CT 601 topics not in the ENCT 352 syllabus

These were asked often in the old syllabus but do not appear in the new one. They are listed for completeness only.

A. Real-time systems

1. ★ Differentiate hard and soft real-time systems. Why is continuous monitoring and control needed in real-time systems? **[2083 Baishakh, 2081 Bhadra, 2080 Bhadra, 2080 Baishakh, 2076 Chaitra, 2076 Ashwin, 2074 Chaitra (short note)]**
2. ★ What is a real-time system? What is the role/use of a data acquisition system (DAS)? Explain with its architecture. **[2082 Bhadra, 2082 Baishakh, 2080 Bhadra, 2078 Kartik, 2078 Bhadra, 2076 Chaitra, 2075 Chaitra, 2071 Chaitra]**
3. How is real-time software different from other / non-real-time software? **[2078 Kartik, 2078 Bhadra, 2075 Chaitra]**

4. Differentiate periodic and aperiodic stimuli. **[2082 Baishakh, 2079 Bhadra]**
5. ★ What is an RTOS? Explain its role, components and characteristics. **[2081 Bhadra, 2081 Baishakh, 2080 Baishakh, 2073 Shrawan, 2068 Chaitra]**
6. ★ How do real-time OS and software differ from non-real-time OS and software? **[2075 Ashwin (short note), 2072 Kartik, 2072 Chaitra, 2071 Chaitra, 2069 Chaitra]**
7. Compare real-time and batch operating systems. **[2073 Chaitra]**
8. Outline the real-time system design process / considerations for designing real-time software. **[2079 Bhadra, 2076 Ashwin, 2071 Shrawan, 2069 Ashad]**
9. Compare DBMS and data acquisition system. **[2070 Ashad]**

B. Software reuse and CBSE

10. ★ What are the benefits and problems/limitations/risks of software reuse? **[2083 Baishakh, 2082 Bhadra, 2082 Baishakh, 2081 Bhadra, 2081 Baishakh, 2080 Bhadra, 2079 Bhadra, 2078 Kartik, 2078 Bhadra, 2076 Chaitra, 2075 Chaitra, 2074 Ashwin, 2068 Chaitra]**
11. ★ What factors must be considered during software reuse planning / before reusing components? **[2082 Bhadra, 2082 Baishakh, 2081 Bhadra, 2080 Baishakh, 2078 Kartik, 2076 Chaitra, 2075 Chaitra, 2074 Ashwin, 2068 Chaitra]**
12. What technical and non-technical factors hinder software reuse? Do you suggest reusing much software? **[2075 Ashwin]**
13. Justify "Advantages of reuse are lower costs, faster development and lower risks." Do you agree that design should be based on systematic reuse? **[2076 Ashwin, 2073 Shrawan, 2069 Ashad]**
14. In which situations is reuse recommended? List the levels at which reuse is possible. **[2078 Bhadra, 2072 Chaitra]**
15. What is a design pattern? Write notes on design patterns and their benefits. **[2076 Ashwin, 2072 Chaitra, 2071 Shrawan]**
16. What is COTS reuse / COTS integration? **[2080 Baishakh, 2079 Bhadra, 2075 Chaitra, 2078 Kartik (short note)]**
17. How useful are application frameworks? Compare reuse framework vs pattern generator and application framework vs component reuse. **[2081 Baishakh, 2070 Ashad, 2069 Chaitra, 2074 Ashwin (short note: pattern generator), 2071 Chaitra (short note: generator based reuse)]**
18. "CBSE is a reuse-based approach to defining and implementing loosely coupled components." Justify. **[2082 Baishakh, 2081 Bhadra, 2073 Chaitra]**
19. ★ What is a component? Explain the CBSE process (for reuse and with reuse). What are its benefits? **[2081 Baishakh, 2080 Baishakh, 2078 Bhadra, 2076 Ashwin, 2074 Chaitra, 2072 Chaitra, 2071 Chaitra, short notes: 2078 Kartik, 2075 Ashwin, 2073 Shrawan, 2071 Shrawan, 2070 Ashad, 2068 Chaitra]**
20. How does CBSE differ from traditional development? **[2080 Bhadra]**
21. Explain provided and required interfaces of a software component with an example. **[2083 Baishakh, 2082 Bhadra, 2079 Bhadra]**
22. What is component composition? Short note on adaptor components. **[2076 Chaitra, 2069 Ashad, 2073 Chaitra]**

C. Cost and size estimation

23. ★ Calculate COCOMO effort, development time, average staffing (and productivity) for:
- 55,000 LOC application program **[2083 Baishakh]**
 - 300 KLOC **[2082 Bhadra]**
 - 200 KLOC semi-detached (COCOMO-2) **[2072 Kartik]**
 - 49,200 LOC application program **[2071 Chaitra]**
 - the TMS / cheque case-study systems **[2075 Ashwin, 2074 Chaitra]**
24. ★ What is COCOMO? Explain its types and use. **[2081 Baishakh, 2078 Bhadra, 2076 Chaitra, 2076 Ashwin, 2071 Shrawan, short notes: 2074 Ashwin, 2073 Shrawan, 2068 Chaitra]**
25. Compute function points (FP) from given counts and weights. **[2082 Baishakh, 2081 Bhadra, 2074 Chaitra]**
26. Explain cost estimation with function-point and object-point methods. List software cost estimation techniques. **[2080 Baishakh, 2074 Ashwin]**
27. Advantages and disadvantages of LOC and FP as size metrics. What is the problem with LOC? What are the size estimation metrics? **[2078 Bhadra, 2073 Chaitra, 2072 Kartik, 2072 Chaitra]**
28. Compute Halstead's metrics (program length, vocabulary, volume, level, effort, time) for given code. **[2075 Chaitra]**

D. CASE tools

29. Why does software engineering need CASE tools? What are their benefits? How is the code generation of CASE tools related to CBSE? **[2082 Baishakh, 2071 Chaitra, 2070 Ashad]**